

Statistical methods for computer science

Alessandro Cardinali

Linear Regression

Introduction

- A regression model is a simple abstraction of reality in the sense that it provides a *relatively simple approximation of some relatively complex phenomenon*
- Models may be broadly classified as *deterministic* or *probabilistic*
- In a *deterministic model*, the outcome of interest is *precisely* defined often by means of a mathematical equation
- Deterministic models abound in the sciences and engineering
- In *probabilistic models*, the outcomes of interest exhibit *variability*, because the model either contains random elements or is impacted in some way by random forces
- There is certainly no more important class of probabilistic model than the probabilistic *regression model*

How does a probabilistic regression model look like?

$$Y = f(x_1, \dots, x_p) + \epsilon \quad (1)$$

- Y is the *outcome of interest* summarizing the phenomenon of interest
- It is also known as *response variable* or *dependent variable*
- $x_1, x_2, \dots, x_p$ are a set of *predictors* or *explanatory variables* or *independent variables*
- $f(x_1, \dots, x_p)$ denotes the *systematic component* of the model
- ϵ denotes *random error term*

Note

- The response variable appearing in eq.(1) is in capital letter; that is, it represents a random variable
- On the other hand, the explanatory variables are in small letters; that is, they are assumed to be observed (fixed) quantities

Some notation

- Let us assume, both the response variable Y and the explanatory variables have been observed for a sample of n statistical units
 - $y_1, \dots, y_n$ denote the observed responses
 - $x_{i1}, \dots, x_{ip}$ denote the observed explanatory variables for the i -th unit in the sample, $i = 1, \dots, n$
- $\mathbf{Y} = (Y_1, \dots, Y_n)' \rightarrow$ vector of random responses
- $\mathbf{y} = (y_1, \dots, y_n)' \rightarrow$ vector of observed responses

Linear regression model

- Within the class of probabilistic regression models, the *linear regression model* represents the most important one
- A linear regression model is defined as

$$Y = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p + \epsilon$$

- Y denotes a *continuous response variable*
- $x_1, x_2, \dots, x_p$ are a set of *predictors* or *explanatory variables*
- $\beta_0, \beta_1, \dots, \beta_p$ are a set of *regression parameters*
- ϵ is *the random error term*

Linear regression model (2)

- For a generic unit i , the linear regression model is specified by

$$Y_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip} + \epsilon_i$$

The response variable is therefore described as the sum of two components

- The systematic component for the i -th unit is defined as a *linear combination* of the explanatory variables

$$f(x_{i1}, \dots, x_{ip}) = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip}$$

This quantity is typically known as *linear predictor*

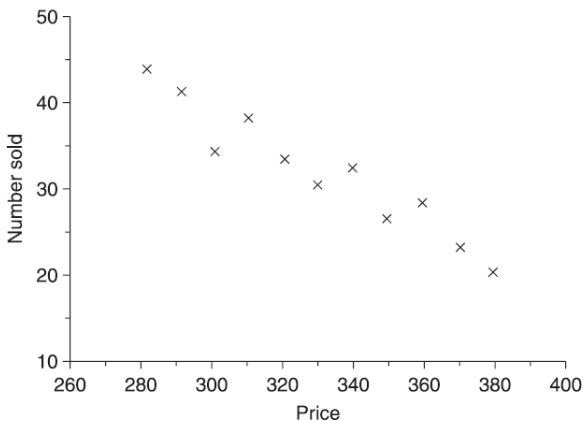
- The error term is meant to capture the variability of the response which cannot be explained in terms of the x 's

■ Example

A new type of washing machine was recently introduced in 11 department stores. These stores are of roughly equal size and are located in similar types of communities. The manufacturer varied the price charged in each store, and the following data, giving the number of units sold in 1 month for each of the different prices, resulted.

Price (\$)	Number sold
280	44
290	41
300	34
310	38
320	33
330	30
340	32
350	26
360	28
370	23
380	20

A plot of the number of units sold y versus the price x for these 11 data pairs is given in Fig. 12.1. The resulting scatter diagram indicates that, subject to random error, the assumption of a straight-line relationship between the number of units sold and the price appears to be reasonable. That is, a simple linear regression model appears to be appropriate.



Vector formulation

- Let $\mathbf{x}_i = (1, x_{i1}, \dots, x_{ip})'$
- The above vector is typically known as the *design vector* for unit i
- Let $\boldsymbol{\beta} = (\beta_0, \beta_1, \dots, \beta_p)'$
- This latter represents the vector of regression parameters

The linear regression model can be written in **vector formulation** as

$$Y_i = \mathbf{x}_i' \boldsymbol{\beta} + \epsilon_i$$

That is

$$Y_i = \underbrace{[1, x_{i1}, \dots, x_{ip}]}_{\text{design vector}} \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_p \end{bmatrix} + \epsilon_i$$

Matrix formulation

- Let $\mathbf{Y} = (Y_1, \dots, Y_n)'$ denote the vector of responses for all units in the sample
- Let $\mathbf{X} = (\mathbf{1}, \mathbf{x}_1, \dots, \mathbf{x}_n)'$ be the matrix of explanatory variables for all units in the sample
- $\mathbf{X}$ is a *full rank matrix* and is typically known as *design matrix*
- Let $\boldsymbol{\epsilon} = (\epsilon_1, \dots, \epsilon_n)$

The linear regression model can be written in **matrix formulation** as

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$$

that is

$$\begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & x_{11} & \cdots & x_{1p} \\ 1 & x_{21} & \cdots & x_{2p} \\ \vdots & \vdots & \cdots & \vdots \\ 1 & x_{n1} & \cdots & x_{np} \end{bmatrix}}_{\text{design matrix}} \times \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_p \end{bmatrix} + \begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_n \end{bmatrix}$$

General and Linear: why?

Frequently, the linear regression model is also called *general linear model*

- **General** refers to the dependence of the response variable on potentially *more than one explanatory variable*
- When only a single explanatory variable is considered as in the example above, the model is referred to as *simple linear regression model*

$$Y_i = \beta_0 + \beta_1 x_i + \epsilon_i.$$

- **Linear** refers to the linearity in the parameters:

$$Y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2}^2 + \epsilon_i$$

$$Y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 e^{x_{i2}} + \epsilon_i$$

but not e.g.

$$Y_i = \beta_0 + \beta_1 x_{i1} + x_{i2}^{\beta_2} + \epsilon_i$$

$$Y_i = \beta_0 + \beta_1 x_{i1} + e^{\beta_2 x_{i2}} + \epsilon_i$$

Model assumptions

The linear regression model relies of some important *assumptions*

- The expected value of ϵ_i is zero

$$E(\epsilon_i) = 0$$

That is, *error terms have zero mean*

- The variance of ϵ_i does not change with i

$$\text{Var}(\epsilon_i) = \sigma^2$$

That is, *error terms have constant variance* equal to $\sigma^2 \rightarrow$
homoschedasticity assumption

- The covariance between error terms associated to two different units is zero

$$\text{Cov}(\epsilon_i, \epsilon_j) = 0$$

That is, *error terms from different units are uncorrelated*

Model assumptions (2)

Being the response variable a function of ϵ , assumptions above directly transfer to the distribution of Y_i

- Expected value of Y_i

$$E(Y_i) = E(\mathbf{x}'_i\boldsymbol{\beta} + \epsilon_i) = \mathbf{x}'_i\boldsymbol{\beta} + E(\epsilon_i) = \mathbf{x}'_i\boldsymbol{\beta} + 0 = \mathbf{x}'_i\boldsymbol{\beta}$$

- The variance of Y

$$\text{Var}(Y_i) = \text{Var}(\mathbf{x}'_i\boldsymbol{\beta} + \epsilon_i) = \text{Var}(\mathbf{x}'_i\boldsymbol{\beta}) + \text{Var}(\epsilon_i) = 0 + \sigma^2 = \sigma^2$$

- The covariance between Y_i and Y_j

$$\text{Cov}(Y_i, Y_j) = \text{Cov}[(\mathbf{x}'_i\boldsymbol{\beta} + \epsilon_i), (\mathbf{x}'_j\boldsymbol{\beta} + \epsilon_j)] = \text{Cov}(\epsilon_i, \epsilon_j) = 0$$

Model assumptions (3)

Based on the above assumptions, we may conclude that

- the expected responses are modeled as a function of observed explanatory variables

$$E(Y_i) = \mu_i = \mathbf{x}_i' \boldsymbol{\beta} \quad (2)$$

- response variables are assumed to have constant variance

$$\text{Var}(Y_i) = \sigma^2$$

- responses from different sample units are uncorrelated each other

$$\text{Cov}(Y_i, Y_j) = 0$$

The above assumptions can be summarized in by using matrix notation as

$$E(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta} = \begin{bmatrix} \mathbf{x}_1\boldsymbol{\beta} \\ \mathbf{x}_2\boldsymbol{\beta} \\ \vdots \\ \mathbf{x}_n\boldsymbol{\beta} \end{bmatrix}$$

$$\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I} = \begin{bmatrix} \sigma^2 & 0 \dots & 0 \\ 0 & \sigma^2 \dots & 0 \\ \vdots & \vdots & \vdots \\ 0 & 0 & \sigma^2 \end{bmatrix}$$

How do we interpret model parameters?

Let us start from a simple linear model (1 explanatory variable)

$$E(Y_i) = \beta_0 + \beta_1 x_i$$

- β_0 is known as *intercept* and represents the mean value of the response when $x_i = 0$

$$E(Y_i) = \beta_0 + \beta_1 \times 0 = \beta_0$$

- *Warning:* The intercept has not practical interpretation when 0 is out of the range of observed values of x and/or is not a reasonable value for x

How do we interpret parameters? (2)

$$E(Y_i) = \beta_0 + \beta_1 x_i$$

To interpret β_1 , let us consider two units with explanatory variables x_i and $x_j = x_i + 1$

$$E(Y_i) = \beta_0 + \beta_1 x_i$$

$$E(Y_j) = \beta_0 + \beta_1 (x_i + 1)$$

Then

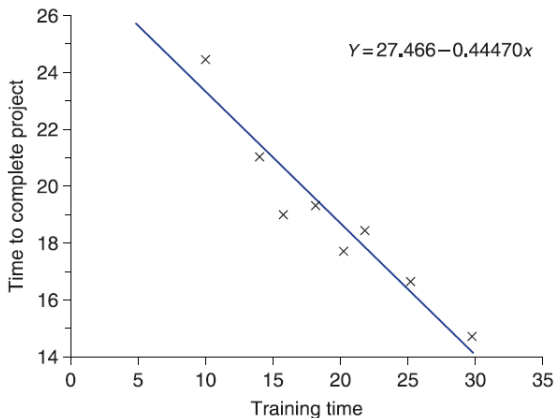
$$\Delta E(Y) = E(Y_j) - E(Y_i) = [\beta_0 + \beta_1 (x_i + 1)] - [\beta_0 + \beta_1 x_i] = \beta_1$$

That is, β_1 denotes the variation in the mean response of Y when the explanatory variable x increases by one unit

■ Example 12.2

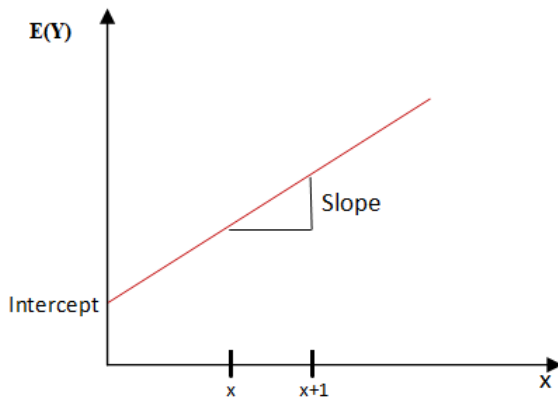
A large midwestern bank is planning on introducing a new word processing system to its secretarial staff. To learn about the amount of training that is needed to effectively implement the new system, the bank chose eight employees of roughly equal skill. These workers were trained for different amounts of time and were then individually put to work on a given project. The following data indicate the training times and the resulting times (both in hours) that it took each worker to complete the project.

Worker	Training time (= x)	Time to complete project (= Y)
1	22	18.4
2	18	19.2
3	30	14.5
4	16	19.0
5	25	16.6
6	20	17.7
7	10	24.4
8	14	21.0



- The intercept has not practical interpretation as 0 is out of the range of x
- When the training time increases by one hour, the time to complete the project reduces by 0.44 hours (about 26 minutes)

How to interpret model parameters? (4)



How do we interpret parameters? (3)

In the case of general linear model (more than one explanatory variable)

$$E(Y_i) = \beta_0 + \beta_1 x_{i1} + \dots + \beta_p x_{ip}$$

Let us consider two units i and j having equal explanatory variables

$$x_{ir} = x_{jr}, \quad r = 1, \dots, p$$

but for the ℓ -th one, for which we have $x_{i\ell}$ and $x_{j\ell} = x_{i\ell} + 1$ Then

$$E(Y_i) = \beta_0 + \beta_1 x_{i1} + \dots + \beta_\ell x_{i\ell} + \dots + \beta_p x_{ip}$$

$$E(Y_j) = \beta_0 + \beta_1 x_{i1} + \dots + \beta_\ell (x_{i\ell} + 1) + \dots + \beta_p x_{ip}$$

$$\Delta E(Y) = E(Y_j) - E(Y_i) = \beta_\ell$$

That is, β_ℓ denotes the variation in the mean response of Y when the ℓ -th explanatory variables increases by one unit, *keeping all other fixed*

Categorical explanatory variables

- The above interpretation clearly holds when the explanatory variables are quantitative
- When dealing with *qualitative* predictors, we cannot define the model as before
- Let us consider a quantitative explanatory variable x and a qualitative explanatory variable w

$$w = \begin{cases} 1 \rightarrow \text{South} \\ 2 \rightarrow \text{West} \\ 3 \rightarrow \text{North} \end{cases}$$

- Defining the regression model as

$$Y_i = \beta_0 + \beta_1 x_i + \beta_2 w_i + \epsilon_i$$

would be misleading

- How do we interpret a parameter e.g. $\beta_2 = 12$?
- The problem is that we cannot interpret the distance between categories of w in a reasonable way

Categorical explanatory variables (2)

A remedy is to define two new explanatory variables, called *dummy* variables, and include them in the model

$$w_1 = \begin{cases} 1 & \text{West} \\ 0 & \text{else} \end{cases}$$

$$w_2 = \begin{cases} 1 & \text{North} \\ 0 & \text{else} \end{cases}$$

NOTE: if k is the number of possible values of w , then $k - 1$ dummy variables are enough to summarize information

$$(w_1, w_2) = (0, 0) \rightarrow \text{South}$$

$$(w_1, w_2) = (1, 0) \rightarrow \text{West}$$

$$(w_1, w_2) = (0, 1) \rightarrow \text{North}$$

Categorical explanatory variables (3)

Then, the model becomes

$$E(Y_i) = \beta_0 + \beta_1 x_i + \beta_2 w_{i1} + \beta_3 w_{i2}$$

- When $x_i = 0$, $w_{i1} = 0$, and $w_{i2} = 0$, we have

$$E(Y_i) = \beta_0$$

β_0 is the mean value of Y when $x_i = 0$ and $w_i = \textit{South}$ (the *reference category*)

- When $x_i = 0$, $w_{i1} = 1$, and $w_{i2} = 0$, we have

$$E(Y_i) = \beta_0 + \beta_2$$

that is, $\beta_0 + \beta_2$ is the mean value of Y when $x_i = 0$ and $w_i = \textit{West}$

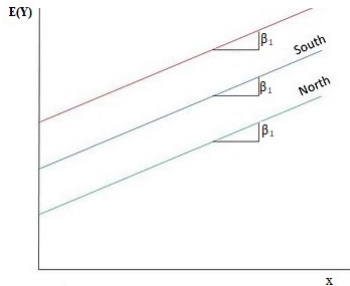
- When $x_i = 0$, $w_{i1} = 0$, and $w_{i2} = 1$, we have

$$E(Y_i) = \beta_0 + \beta_3$$

that is, $\beta_0 + \beta_3$ is the mean value of Y when $x_i = 0$ and $w_i = \textit{North}$

- β_1 is the variation in $E(Y)$ when x_i increases by one unit, keeping w_{i1} and w_{i2} constant

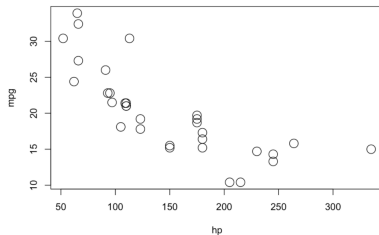
Categorical explanatory variables (4)



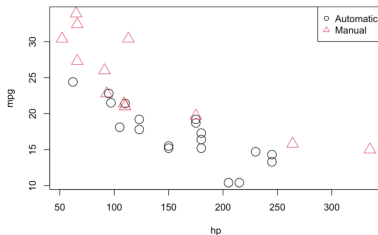
An example

- Suppose, the response we are interested in is fuel efficiency of cars (*mpg*)
- Suppose the value of such a response is observed on a sample of 32 cars together with the following predictors
 - *hp*: horsepower of the car
 - *am*: transmission $\rightarrow$ manual or automatic

Suppose, we are interested in mpg as the response variable, and hp as a predictor

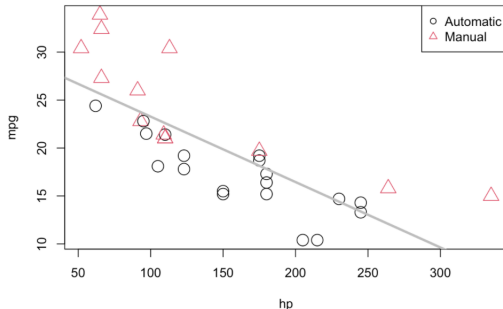


Suppose, we are also interested in the transmission type. We can color the data points on the plot accordingly



Let us consider the following regression model

$$mpg_i = \beta_0 + \beta_1 * hp + \epsilon_i$$



where the values of β_0 and β_1 are obtained according a procedure we will detail later on.

By the way, looking at the graph, it is evident that the red points, manual observations, largely fall above the line, while the black, automatic observations, are mostly below the line

To correct for this, we can add a further predictor to our model, by considering a single dummy variable

$$manual_i = \begin{cases} 1 & \text{manual transmission} \\ 0 & \text{automatic transmission} \end{cases}$$

The model becomes therefore

$$mpg_i = \beta_0 + \beta_1 * hp + \beta_2 * manual_i + \epsilon_i$$

Since the variable *manual* can only take values 0 and 1, we can effectively write two different models

- one for automatic transmissions

$$mpg_i = \beta_0 + \beta_1 * hp + \epsilon_i$$

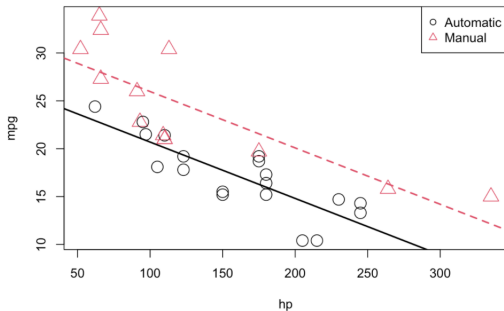
since the dummy variable *manual* = 0

- one for manual transmissions

$$mpg_i = (\beta_0 + \beta_2) + \beta_1 * hp + \epsilon_i$$

since the dummy variable *manual* = 1

- Note that the two models share the same slope, β_1 , but have different intercepts, differing by β_2
- So, the change in mpg is the same for both models, but on average mpg differs by β_2 between the two transmission types



Note that the points are no longer systematically incorrect. The red/black observations, vary about the red/black line in no particular pattern as before

A more complex example

- Let us assume, we are interested in the predictor *cyl* identifying the number of cylinders the car owns
- Three possible values for the predictors are observed: 4, 6, and 8 cylinders
- It doesn't make sense assume the difference in the mean mpg between 4 and 6 cylinders is the same as the difference in the mean mpg between 6 and 8 cylinders
- Therefore, we may not consider *cyl* as a quantitative attribute, but rather as a qualitative one with three categories

- We can define two dummy variables

$$v_1 = \begin{cases} 1 & \text{if 6 cylinders} \\ 0 & \text{else} \end{cases}$$

$$v_2 = \begin{cases} 1 & \text{if 8 cylinders} \\ 0 & \text{else} \end{cases}$$

- Let us consider now the regression model

$$Y_i = \beta_0 + \beta_1 * hp + \beta_2 * v_1 + \beta_3 * v_2 + \epsilon_i$$

$$Y_i = \beta_0 + \beta_1 * hp + \beta_2 * v_1 + \beta_3 * v_2 + \epsilon_i$$

Also in this case, we can effectively write three different models

- one for 4 cylinders

$$Y_i = \beta_0 + \beta_1 * hp + \epsilon_i$$

- one for 6 cylinders

$$Y_i = (\beta_0 + \beta_2) + \beta_1 * hp + \epsilon_i$$

- one for 8 cylinders

$$Y_i = (\beta_0 + \beta_3) + \beta_1 * hp + \epsilon_i$$

The *method of ordinary least squares (OLS)* is typically used to estimate the regression parameters in a general linear regression model

$$Y_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip} + \epsilon_i = \beta_0 + \sum_{k=1}^p \beta_k x_{ik} + \epsilon_i \quad (3)$$

OLS chooses the β 's in equation (3) so that to ensure the best fit to the observed data

Ordinary least square (OLS) estimation

A measure of coherence between the observed data (y_i) and the model ($\mathbf{X}\beta$) is given by the sum of the squared errors. That is

$$S = \sum_{i=1}^n \epsilon_i^2 = \sum_{i=1}^n \left(y_i - \beta_0 - \sum_{k=1}^p \beta_k x_{ik} \right)^2$$

- The function S is to be minimized with respect to $\beta_0, \beta_1, \dots, \beta_p$
- The *least squares estimators*, say, $\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_p$ must satisfy

$$\left. \frac{\partial S}{\partial \beta_0} \right|_{\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_p} = -2 \sum_{i=1}^n \left(y_i - \hat{\beta}_0 - \sum_{k=1}^p \hat{\beta}_k x_{ik} \right) = 0$$

and

$$\left. \frac{\partial S}{\partial \beta_k} \right|_{\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_p} = -2 \sum_{i=1}^n \left(y_i - \hat{\beta}_0 - \sum_{k=1}^p \hat{\beta}_k x_{ik} \right) x_{ik} = 0, \quad k = 1, 2, \dots, p$$

Matrix formulation

- The above equations, are known as *ordinal least square equations*
- The solution to the OLS equations, are known as *ordinary least square estimators*

A simpler way to solve OLS equations is by means of the matrix formulation of the linear regression model

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$$

We wish to find the vector of least squares estimators, $\boldsymbol{\beta}$, that minimizes

$$S(\boldsymbol{\beta}) = \sum_{i=1}^n \epsilon_i^2 = \boldsymbol{\epsilon}'\boldsymbol{\epsilon} = (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})'(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})$$

Matrix formulation (2)

$$S(\beta) = \sum_{i=1}^n \epsilon_i^2 = \epsilon' \epsilon = (\mathbf{y} - \mathbf{X}\beta)'(\mathbf{y} - \mathbf{X}\beta)$$

- Computing the matrix product in the above summation, we get

$$\begin{aligned} S(\beta) &= \mathbf{y}'\mathbf{y} - \beta'\mathbf{X}'\mathbf{y} - \mathbf{y}'\mathbf{X}\beta + \beta'\mathbf{X}'\mathbf{X}\beta \\ &= \mathbf{y}'\mathbf{y} - 2\beta'\mathbf{X}'\mathbf{y} + \beta'\mathbf{X}'\mathbf{X}\beta \end{aligned}$$

since $\beta'\mathbf{X}'\mathbf{y}$ is a scalar, and its transpose $\mathbf{y}'\mathbf{X}\beta = \beta'\mathbf{X}'\mathbf{y}$

- The least square estimators $\hat{\beta}$ must satisfy

$$\left. \frac{S(\beta)}{\beta} \right|_{\hat{\beta}} = -2\mathbf{X}'\mathbf{y} + 2\mathbf{X}'\mathbf{X}\hat{\beta} = \mathbf{0}$$

which simplifies to

$$\mathbf{X}'\mathbf{X}\hat{\beta} = \mathbf{X}'\mathbf{y}$$

Matrix formulation (3)

$$\mathbf{X}'\mathbf{X}\hat{\beta} = \mathbf{X}'\mathbf{y} \quad (4)$$

- The above expression gives the ordinary least squares equations in a matrix form
- As long the design matrix $\mathbf{X}$ is a full rank matrix, $\mathbf{X}'\mathbf{X}$ is *positive definite*
- To solve the OLS equations, we can multiply both sides of (4) by the inverse of $\mathbf{X}'\mathbf{X}$
- The OLS estimator for β is given by

$$\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}$$

The fitted model

The fitted value for the i -th unit in the sample is given by

$$\hat{y}_i = \hat{\beta}_0 + \sum_{k=1}^p \hat{\beta}_k \mathbf{x}_{ik}, \quad i = 1, \dots, n$$

Using a matrix formulation, the vector of fitted values is denoted by

$$\hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}\mathbf{y}$$

The difference between the observed value y_i and the fitted value $\hat{y}_i$ is known as *residual*

$$u_i = y_i - \hat{y}_i = y_i - \hat{\beta}_0 - \sum_{k=1}^p \hat{\beta}_k \mathbf{x}_{ik}$$

Using a matrix formulation, the vector of residuals is denoted by

$$\mathbf{u} = \mathbf{y} - \hat{\mathbf{y}}$$

Properties of $\hat{\beta}$

- An estimator of a parameter is said to be *unbiased* if its expected value is equal to the true parameter value
- The method of least squares produces an *unbiased estimator* of the parameter β in the linear regression model

$$\begin{aligned}E[\hat{\beta}] &= E[(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}] \\&= E[(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'(\mathbf{X}\beta + \epsilon)] \\&= E[(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}\beta + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\epsilon] \\&= E[(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}\beta] + E[(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\epsilon] \\&= \beta + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'E(\epsilon) \\&= \beta\end{aligned}$$

as $E(\epsilon) = 0$ and $(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X} = \mathbf{I}$

- Thus, $\hat{\beta}$ is an unbiased estimator for β

Properties of $\hat{\beta}$ (2)

As regards the variability of $\hat{\beta}$, the covariance matrix can be expressed as

$$\text{Cov}(\hat{\beta}) = E[(\hat{\beta} - E(\hat{\beta}))(\hat{\beta} - E(\hat{\beta}))']$$

The above matrix is a $(p + 1) \times (p + 1)$ dimensional matrix

- the k -th element on the diagonal corresponds to the variance of $\hat{\beta}_k$
- off-diagonal elements correspond to the covariance between pairs $(\hat{\beta}_k, \hat{\beta}_\ell)$

To compute $\text{Cov}(\hat{\beta})$, we may proceed as follows

$$\begin{aligned}\text{Cov}[\hat{\beta}] &= \text{Cov}[(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}] \\ &= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\text{Cov}(\mathbf{y})\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} \\ &= \sigma^2(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} \\ &= \sigma^2(\mathbf{X}'\mathbf{X})^{-1}\end{aligned}$$

Estimation of the error variance

- Once β has been estimated, we need to do the same for σ^2
- To develop an estimator of this parameter, consider the sum of squares of the residuals

$$SS_{res} = \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \sum_{i=1}^n u_i^2 = \mathbf{u}'\mathbf{u}$$

- Substituting $\mathbf{u} = \mathbf{y} - \hat{\mathbf{y}} = \mathbf{y} - \mathbf{X}\hat{\beta}$, we have

$$\begin{aligned} SS_{res} &= (\mathbf{y} - \mathbf{X}\hat{\beta})'(\mathbf{y} - \mathbf{X}\hat{\beta}) \\ &= \mathbf{y}'\mathbf{y} - \hat{\beta}'\mathbf{X}'\mathbf{y} - \mathbf{y}'\mathbf{X}\hat{\beta} + \hat{\beta}'\mathbf{X}'\mathbf{X}\hat{\beta} \\ &= \mathbf{y}'\mathbf{y} - 2\hat{\beta}'\mathbf{X}'\mathbf{y} + \hat{\beta}'\mathbf{X}'\mathbf{X}\hat{\beta} \end{aligned}$$

- Because $\mathbf{X}'\mathbf{X}\hat{\beta} = \mathbf{X}'\mathbf{y}$, the last equation becomes

$$SS_{res} = \mathbf{y}'\mathbf{y} - \hat{\beta}'\mathbf{X}'\mathbf{y}$$

Estimation of the error variance (2)

$$SS_{res} = \mathbf{y}'\mathbf{y} - \hat{\beta}'\mathbf{X}'\mathbf{y}$$

- The above equation is known as residual sum of squares
- It can be shown that

$$E(SS_{res}) = \sigma^2[n - (p + 1)]$$

- An unbiased estimator for σ^2 is given by

$$\hat{\sigma}^2 = \frac{SS_{res}}{n - p - 1}$$

Maximum likelihood estimation (MLE)

- A different approach to estimate model parameters is to look for the vector of parameters $\hat{\beta}$ that makes the observed data *the “most likely”*
- A measure of coherence between the observed data and the model is given by the **likelihood function**

$$L(\beta, \sigma^2 \mid \mathbf{y}) = f(y_1, \dots, y_n \mid \beta, \sigma^2)$$

- $f(y_1, \dots, y_n \mid \beta, \sigma^2)$ denotes the density/distribution function of the vector of observed data $\mathbf{y} = (y_1, \dots, y_n)$, under the assumed regression model

NOTE: to use such a method, we need to specify the distribution for the responses

Gaussian distribution

To derive the ML estimator of β , we add a further hypothesis to the previous ones

- $E(\epsilon) = \mathbf{0}$
- $\text{Cov}(\epsilon) = \sigma^2 \mathbf{I}$
- + $\epsilon \sim N(\mathbf{0}, \sigma^2 \mathbf{I}) \rightarrow$ error terms follow a **Gaussian distribution**

As a result of the previous hypotheses,

$$\mathbf{Y} \sim N(\mathbf{X}\beta, \sigma^2 \mathbf{I})$$

so that

$$f(\mathbf{y} \mid \beta, \sigma^2) = (2\pi)^{-n/2} |\sigma^2 \mathbf{I}|^{-1/2} \exp \left\{ -\frac{1}{2} (\mathbf{y} - \mathbf{X}\beta)' [\sigma^2 \mathbf{I}]^{-1} (\mathbf{y} - \mathbf{X}\beta) \right\}$$

Maximum likelihood estimation

Going back to the problem of estimating the vector of model parameters β , the likelihood function is defined as

$$\begin{aligned} L(\beta, \sigma^2 \mid \mathbf{y}) &= f(\mathbf{y} \mid \beta, \sigma^2) \\ &= (2\pi)^{-n/2} |\sigma^2 \mathbf{I}|^{-1/2} \exp \left\{ -\frac{1}{2} (\mathbf{y} - \mathbf{X}\beta)' [\sigma^2 \mathbf{I}]^{-1} (\mathbf{y} - \mathbf{X}\beta) \right\} \\ &= (2\pi)^{-n/2} \sigma^{-n} \exp \left\{ -\frac{1}{2\sigma^2} (\mathbf{y} - \mathbf{X}\beta)' (\mathbf{y} - \mathbf{X}\beta) \right\} \end{aligned}$$

If we consider the logarithmic transform of the likelihood function (which is easier to maximize), then we have

$$\ell(\beta, \sigma^2 \mid \mathbf{y}) = -\frac{n}{2} \log(2\pi) - n \log \sigma - \frac{1}{2\sigma^2} (\mathbf{y} - \mathbf{X}\beta)' (\mathbf{y} - \mathbf{X}\beta)$$

Maximum likelihood estimation (MLE)

ML estimate of β is derived as

$$\hat{\beta}_{\text{ML}} = \arg \max_{\beta} L(\beta, \sigma^2 \mid \mathbf{y}) \quad (5)$$

$$= \arg \max_{\beta} \ell(\beta, \sigma^2 \mid \mathbf{y}) \quad (6)$$

$$= \arg \min_{\beta} (\mathbf{y} - \mathbf{X}\beta)'(\mathbf{y} - \mathbf{X}\beta) \quad (7)$$

Estimating model parameters via MLE and OLS leads to identical results

$$\hat{\beta} = \hat{\beta}_{\text{ML}} = \hat{\beta}_{\text{OLS}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}$$

Goodness-of-fit measure

Total variability in $Y \rightarrow$ total sum of squares

$$SS_T = \sum_{i=1}^n (y_i - \bar{y})^2$$

TSS can be expressed as the sum of two components

- *explained sum of squares*

$$SS_R = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2$$

- *residual sum of squares*

$$SS_{res} = \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \sum_{i=1}^n u_i^2$$

Goodness-of-fit measure (II)

Starting from

$$SS_T = SS_R + SS_{res}$$

we may define a goodness-of-fit measure for the general linear model as

$$R^2 = \frac{SS_R}{SS_T} = 1 - \frac{SS_{res}}{SS_T}$$

- $0 \leq R^2 \leq 1$ is known as the **coefficient of multiple determination**
- The goodness of fit increases with R^2

The coefficient of determination can be directly interpreted as the proportion of total variability of Y explained by the regression model

- Every time a predictor is added to the model, the **R-squared cannot decrease**, regardless of whether the predictor has or has not an impact on Y
- Consequently, a model with more terms may appear to have a better fit simply because it has more terms
- If a model has too many predictors, it begins model the random noise in the data (**overfitting**)

Adjusted R^2 (2)

For the above reasons, we may rely on a modified version of the R^2 which takes into account the number of predictors in the model

$$R_{adj}^2 = 1 - \frac{SS_{res}/(n - p - 1)}{SS_T/(n - 1)} = 1 - \left(\frac{n - 1}{n - p - 1} \right) (1 - R^2)$$

- The adjusted R^2 does not always increase when variables are added to the model
- If such variables are unnecessary, the adjusted R^2 decreases
- This index ranges in the same interval as R^2
- However, it cannot be directly interpreted in terms of the proportion of explained variability of the response

Inference on model parameters

At the basis of ML estimation, we have the assumption on the distribution of the responses in $\mathbf{Y}$

$$\mathbf{Y} \sim N(\mathbf{X}\beta, \sigma^2 \mathbf{I})$$

- Thanks to this assumption, we can make **inference** on model parameters
- That is, we can extend the results obtained from the observed data to the whole population
- Among the inferential procedures, hypothesis testing represents one of the most useful

Hypothesis testing

- A **significance test** is an inferential method that uses sample data to summarize the evidence about a hypothesis
- In other terms, it is a method to check whether or not the data support certain statements or predictions
- A significance test about a given hypothesis has five steps
 - Assumptions
 - Hypotheses
 - Test Statistic
 - p-value
 - Conclusion

Hypothesis testing (II)

Assumptions

- Each significance test is based upon certain assumptions about some characteristics of the population

$$\mathbf{Y} \sim N(\mathbf{X}\beta, \sigma^2 \mathbf{I})$$

- Being $\hat{\beta}$ a linear combination of observed responses, than the estimator $\hat{\beta}$ inherits the same distribution as $\mathbf{Y}$

$$\hat{\beta} \sim N(\beta, \sigma^2 (\mathbf{X}'\mathbf{X})^{-1})$$

or equivalently

$$\hat{\beta}_k \sim N(\beta_k, \sigma^2 h_{kk})$$

where h_{kk} denotes the k -th diagonal element of $(\mathbf{X}'\mathbf{X})^{-1}$

Hypothesis testing (III)

System of hypothesis

A hypothesis is a statement about a population, usually of the form that a certain parameter takes a particular numerical value. Each significance test has two hypotheses

- The **null hypothesis** (H_0) states that the parameter of interest takes a particular value (a single value)
- The **alternative hypothesis** H_1 states that the parameter falls in some alternative range of values. It specifies how the null can be false

In the regression framework, we frequently have

$$H_0 : \beta_k = 0$$

$$H_1 : \beta_k \neq 0$$

Hypothesis testing (IV)

Test statistic

- In order to decide whether the null should be rejected or not, we summarize the evidence against the null by computing a **test statistic**
- A test statistic describes (measures) how far the point estimate falls from the parameter value given in the null hypothesis
- The test statistics is zero when the estimate is equal to the value specified in the null
- If the test statistic falls far from zero in the direction specified by the alternative hypothesis, this is interpreted as evidence against the null hypothesis and in favor of the alternative one

Hypothesis testing (IV)

Test statistic

In the regression framework, given the hypothesis

$$H_0 : \beta_k = 0$$

$$H_1 : \beta_k \neq 0$$

the test statistic is given by

$$Z = \frac{\hat{\beta}_k - 0}{\sqrt{\sigma^2 h_{kk}}}$$

Hypothesis testing (V)

p-value

To interpret a test statistic value, we use a probability summary of the evidence against the null hypothesis H_0

- First, we presume that H_0 is true
- Next, we consider the sampling distribution of the test statistic under such a hypothesis

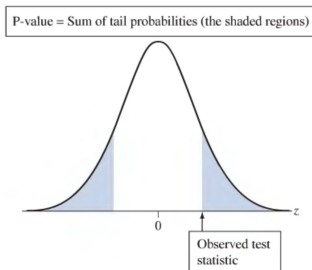
$$Z = \frac{\hat{\beta}_k - 0}{\sqrt{\sigma^2 h_{kk}}} \sim N(0, 1)$$

We summarize how far in the tail of this distribution the test statistic falls by computing, when H_0 is true, the probability of observing a value of Z more extreme than the observed one

$$\Pr(|Z| > |z_{obs}|) = \begin{cases} Z > z_{obs} \\ Z < -z_{obs} \end{cases}$$

This probability is called **p-value**

Hypothesis testing (VI)



The smaller the p-value, the stronger the evidence the data provide against the null hypothesis

That is, a small p-value indicates a small likelihood of observing the sampled results if the null hypothesis was true

Hypothesis testing (VII)

Starting from the p-value

$$\Pr(|Z| > |z_{obs}|) = \begin{cases} Z > z_{obs} \\ Z < -z_{obs} \end{cases}$$

Conclusions

- We reject H_0 if p-value $<$ a given significance level α
- We do not reject H_0 if p-value $>$ a given significance level α
- Typically, we fix $\alpha = 0.05$

Hypothesis testing (VIII)

The error variance is unknown and is estimated by $\hat{\sigma}^2 = \frac{\mathbf{u}'\mathbf{u}}{n-p-1}$.
In place of the test statistic Z , we consider

$$T = \frac{\hat{\beta}_k - 0}{\sqrt{\hat{\sigma}^2 h_{kk}}}$$

Under the null hypothesis H_0 , T follows a **Student's t distribution** with $n - p - 1$ **degrees of freedom** (this is very similar to the standard Gaussian distribution, but has fatter tails)

We can then calculate a p-value for the test statistic as

$$\text{p-value} = \Pr(|T| \geq |t_{\text{obs}}|)$$

We reject the null hypothesis if the p-value is smaller than α

Test for all predictors

We can perform a global test of hypothesis, by considering the following **system of hypothesis**

$$\begin{cases} H_0 : \beta_1 = \dots = \beta_k = \dots = \beta_p = 0 \\ H_1 : \beta_k \neq 0 \text{ for at least one } k = 1, \dots, p \end{cases}$$

The **test statistic** to use in this case is

$$F = \frac{SS_R/p}{SS_{res}/(n - p - 1)}$$

Under H_0 , F follows a **Fisher's F distribution** with p and $n - p - 1$ degrees of freedom

We can then calculate the p-value for the test statistic as

$$p - \text{value} = \Pr(F \geq f_{obs} \mid H_0)$$

We reject the null hypothesis if the p-value is smaller than α